

Sample Questions for Class Test II CITS2200 Data Structures and Algorithms

Q1. Which of the following statements is false for a doubly-linked list? Assume that you are given the reference **current** of an arbitrary node in the list, and each node stores two references **next** and **previous** to the node to its right and left

(a) Finding the **next** and **previous** references in a doubly-linked list is constant-time operations; (b) A doubly-linked list is more space-efficient compared to a singly-linked list ; (c) The complexity of finding the reference of the last node in the list is $O(n)$; (d) Inserting a new node between the references **previous** and **current** takes $O(1)$ time.

Q2. Which of the following statements is false?

(a) The time complexity of **selection sort** is $O(n^2)$; (b) The time-complexity of **insertion sort** is $O(n^2)$; (c) The time-complexity of **heap sort** is $O(n^2)$; (d) **Heap sort** is an optimal sorting algorithm.

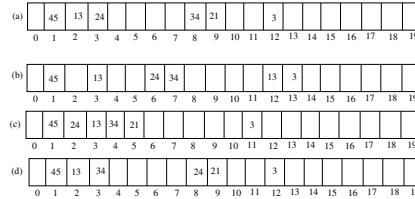
Q3. Which of the following statements is true for a **min-heap**?

(a) All the values in a subtree of a node k are smaller than the value at node k ; (b) The worst-case time complexity of restoring the heap property after inserting a new value is $O(n)$.; (c) The values on a root to leaf path in the heap is in ascending order; (d) The worst-case time complexity of constructing a heap, given n integers is $O(n \log n)$.

Q4. Which of the followings is a valid representation of a single binary tree in an array. Assume that the root is stored in the location with array index 1.

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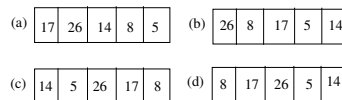
Q5. We now delete the number 7 from the binary tree in Q4. Which of the following is a valid representation of the binary tree now?



Q6. Each node of a singly-linked list can store an `item` and a reference `next` to the node to the right. There are three nodes A, B and C in a linked-list such that $A.next=B$ and $B.next=C$. Which of the following deletes the node B correctly?

- (a) $A.next=B$ (b) $A.next=A.next.next$ (c) $A.next=B$ (d) $B.next=A$
 $B.Next=C$

Q7. You are storing integers 26, 17, 14, 8, 5 in an array of size 5, using a hash function $(N + 6) \bmod 5$ when N is the integer you are storing. Which of the following is a correct representation of the array after storing these integers? The array index starts from 0.



Q8. You are maintaining a priority queue using a linked list. You insert every element in the correct sorted position in the list. What are the worst-case times for enqueue and dequeue operations from this priority queue?

(a) $O(1)$ and $O(n)$; (b) $O(n)$ and $O(\log n)$; (c) $O(n)$ and $O(1)$; (d) none of the above.

Q9. You are given two min-heaps of n nodes each. You want to merge these two min-heaps by adding a root node and making the roots of the two

min-heaps as children of this new root node. What is the time complexity of merging these two min-heaps?

(a) $O(n \log n)$; (b) $O(n)$; (c) $O(\log n)$; (d) $O(n^2)$.

Q10. Given an unsorted array of n numbers, what is the complexity of the best algorithm using a heap for finding the minimum number in the array?

(a) $O(\log n)$; (b) $O(n \log n)$; (c) $O(n^2)$; (d) $O(n)$.